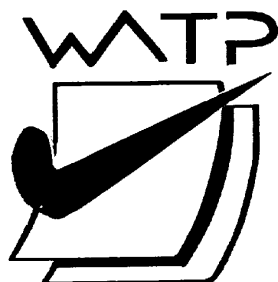


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SEMESTER ONE

**MATHEMATICS
METHODS
UNITS 1**

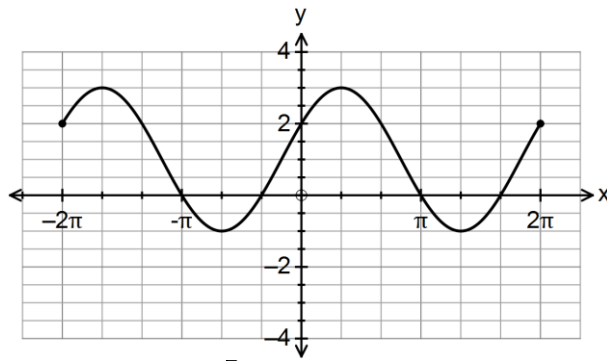
2018

SOLUTIONS

Calculator-free Solutions

1. (a) (i) $98 = a(16) + 44 + 6$
 $a = 3$ ✓
 $g(-2) = -4$ ✓
(ii) $x = -\frac{11}{6}$ ✓
(iii) Minimum ✓
(b) $b^2 - 4ac < 0$
 $121 - 4a(6) < 0$ ✓
 $-24a < -121$
 $a > \frac{121}{24}$ ✓ [6]
2. (a) (i) $y = \tan\left(\frac{x}{2}\right) + 1$ ✓✓
(ii) $x = -y^2 + 2$ ✓✓
(b) $y = \tan\left(\frac{x}{2}\right) + 1$ is a function. For each value of x there is only one corresponding y value. (If a vertical line is drawn anywhere on this function it will only intersect the graph a maximum of once.) ✓✓ [6]
3. (a) Line A: $y = \frac{1}{3}x$
Line B: $y = -3x$ ✓
 $y = -3(2)$ ✓
 $= -6$
(b) Line CD: $-6 = \frac{1}{3}(2) + c$
 $c = -\frac{20}{3} \therefore y = \frac{1}{3}x - \frac{20}{3}$ ✓
Point C(-1, -7) ✓
 $\frac{-1+x}{2} = 2$ $\frac{-7+y}{2} = -6$ ✓
 $x = 5$ and $y = -5$
D(5, -5) ✓ [6]
4. (a) $\frac{4}{5}$ ✓
(b) $\tan \theta = -1$ or $\sin \theta = 0$ or $\sin \theta = 1$ ✓
 $\theta = \frac{3\pi}{4}, \frac{7\pi}{4}, -\frac{\pi}{4}, 0, \pi, 2\pi, -\pi, \frac{\pi}{2}$ ✓✓ [4]
5. $x^2 - \frac{7}{5}x + \frac{1}{5} = 0$
 $\left(x - \frac{7}{10}\right)^2 - \frac{49}{100} + \frac{1}{5} = 0$ ✓
 $x - \frac{7}{10} = \pm \frac{\sqrt{29}}{10}$ ✓
 $x = \frac{7 + \sqrt{29}}{10}$ or $\frac{7 - \sqrt{29}}{10}$ ✓ [3]

6. (a)



- (b) (i) $-\pi, -\frac{\pi}{3}, \pi, \frac{5\pi}{3}$
 (ii) $-\frac{4\pi}{3}, -\frac{2\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}$

✓✓✓

✓✓

✓✓

[7]

7. (a) $\frac{x^3 + 2x^2 - 5x - 6}{x + 1} = x^2 + x - 6$

✓

$$p(x) = (x + 1)(x + 3)(x - 2)$$

✓✓

(b) (i) $y = \frac{1}{2}(x + 1)(x - 2)^2$

✓✓

(ii) Vertical dilation, factor $\frac{1}{2}$.

✓

Horizontal translation one unit to the right

✓

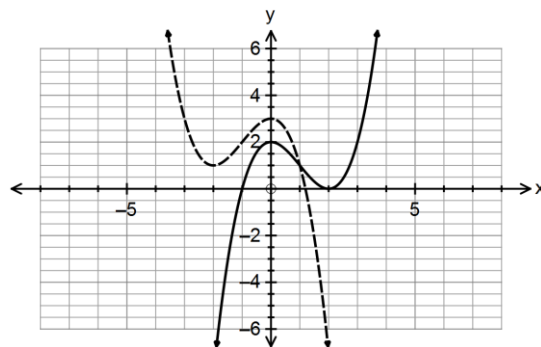
Horizontal dilation, factor $\frac{1}{2}$.

✓

(ii) $\left(\frac{1}{2}, 1\right)$

✓

(iii)



✓✓

[11]

8. (a) (i)

5	1	5	10	10	5	1	
6	1	6	15	20	15	6	1

✓✓

(ii) $p^6 - 6p^5q + 15p^4q^2 - 20p^3q^3 + 15p^2q^4 - 6pq^5 + q^6$

✓✓

(b) $a = 10$ $b = 15$ $c = 10$

✓✓

[6]

9. $\cos\left(\frac{5\pi}{12}\right) = \cos\left(\frac{\pi}{4} + \frac{\pi}{6}\right) = \cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{6}\right) - \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{6}\right)$
 $= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$
 $= \frac{\sqrt{6} - \sqrt{2}}{4}$

✓

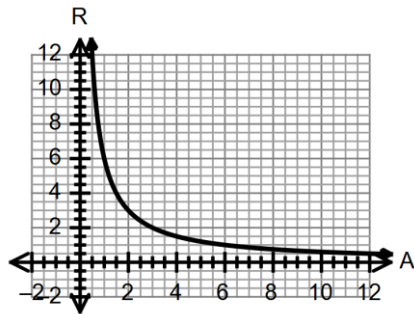
✓

✓

[3]

Calculator-assumed Solutions

10. (a) $k = 6$ ✓
 (b) $R = \frac{6}{5} = 1.2$ ohms ✓
 (c) $R = \frac{6}{A}$ ✓
 $\left(\frac{1}{1.35}\right)R = \frac{6}{1.35A}$ ✓
 R will decrease by 74% ✓
 (d)



✓✓✓

[7]

11. (a)

		Number of toilets			
		1	2	3 or more	Total
Number of bedrooms	1	35	0	0	35
	2	50	5	0	55
	3	5	65	15	85
	4 or more	0	0	25	25
Total		90	70	40	200

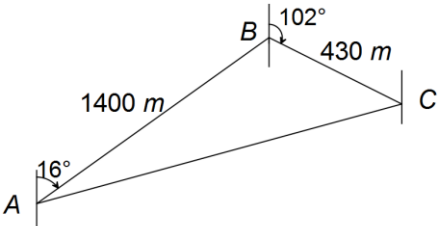
✓✓

- (b) (i) $\frac{85}{200}$ ✓
 (ii) $\frac{110}{200}$ ✓
 (iii) $\frac{65}{200} + \frac{15}{200} = \frac{80}{200}$ ✓✓
 (iv) $\frac{5}{70}$ ✓✓
 (c) $\frac{{}^5C_1 \times {}^{65}C_1 \times {}^{15}C_1}{{}^{85}C_3} = \frac{4875}{98770} = 0.04936$ ✓✓

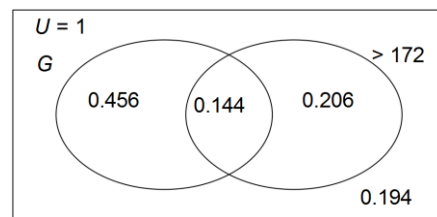
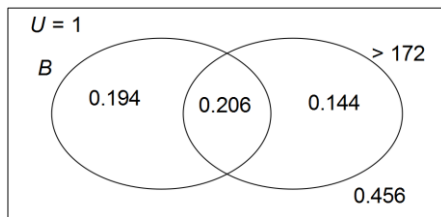
[10]

12. (a) $\widehat{PQ} = r\theta$
 $= 12\pi$ cm ✓✓
 (b) $\frac{\pi}{3}$ ✓
 (c) Area of sector = $\frac{1}{2}r^2\theta$
 $= \left(\frac{1}{2}\right)(18)^2\left(\frac{2\pi}{3}\right)$
 $= 108\pi$ cm² ✓
 $\tan \frac{\pi}{3} = \frac{PR}{18} \rightarrow PR = 18\sqrt{3}$ ✓
 Area of kite = $2\left(\frac{1}{2} \times 18 \times 18\sqrt{3}\right) = 324\sqrt{3}$ cm² ✓
 Area of shaded area = $324\sqrt{3} - 108\pi$ cm² ✓

[7]

13. (a) $(x-5)^2 + (y-8)^2 = 25$ ✓
 $x^2 - 10x + 25 + y^2 - 16y + 64 = 25$ ✓
 $x^2 + y^2 - 10x - 16y + 64 = 0$
- (b) (i) M(2, 4) N(8, 4) ✓
MN = 6 units ✓
- (ii) $6^2 = 5^2 + 5^2 - 50 \cos \theta$ ✓
 $\theta = 1.2870$ ✓
 $A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $A = \frac{1}{2} (5^2) (1.287 - \sin 1.287)$ ✓
 $A = 4.0875 \text{ units}^2$ ✓ [7]
14. (a) (i) $\frac{1}{2}$ ✓
(ii) $\frac{1}{12}$ ✓
(iii) $\frac{2}{12}$ ✓
- (b) $P(A \cap B) = \frac{1}{12} \neq 0$ ✓
- (c) (i) $\frac{2}{12}$ ✓
(ii) $\frac{1}{4}$ ✓
(iii) $\frac{9}{12}$ ✓ [7]
15. (a) 
 $\angle ABC = 94^\circ$ ✓
 $AC^2 = 1400^2 + 430^2 - 2(1400)(430)\cos 94^\circ$ ✓
 $AC = 1492.95 \text{ m}$ ✓
- (b) $\frac{\sin C}{1400} = \frac{\sin 94}{1492.95}$ ✓
 $\angle C = 69.3^\circ$ ✓
 $270^\circ - (69.3 - 12) = 212.7^\circ$ or $180 + (102 - 69.3) = 212.7^\circ$ ✓
Bearing is 212.7°T ✓ [6]

16. (a)



- (b) (i) 0.144
(ii) 0.194
(iii) $\frac{0.144}{0.6} = 0.24$
- (c) Girls over 172 cm = 38
Boys over 172 cm = 55
38:55 = 1:1.45

✓✓✓

✓

✓

✓✓

✓

✓

✓

[10]

17. (a) $\sin\left(\frac{\pi}{6}\right) = \frac{PR}{r} \therefore PR = \frac{r}{2}$
 $\cos\left(\frac{\pi}{6}\right) = \frac{OR}{r} \therefore OR = \frac{\sqrt{3}r}{2}$
 $A = \frac{1}{2} \left(\frac{r}{2} \right) \left(\frac{\sqrt{3}r}{2} \right) = \frac{r^2\sqrt{3}}{8}$
- (b) $A = \frac{1}{2} r^2 \left(\frac{\pi}{6} \right) = \frac{\pi r^2}{12}$
 $\frac{\pi r^2}{12} - \frac{r^2\sqrt{3}}{8} = \frac{2\pi - 3\sqrt{3}}{6}$
 $r = 2 \text{ cm}$

✓

✓

✓

✓

✓

✓

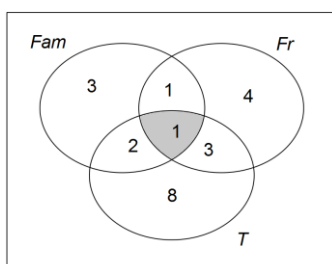
✓

[6]

18. (a) (i) $15! = 1\,307\,674\,368\,000$
(ii) ${}^3C_2 \times {}^4C_2 \times {}^8C_2 = 504$
(iii) $\frac{{}^3C_2 \times {}^4C_2 \times {}^8C_2}{{}^{15}C_6} = \frac{72}{715} = 0.100699$
- (b) (i)

✓✓

✓✓



✓

This image has family and friends and travel in it.

✓

- (ii) $\frac{4}{14}$

✓✓

[9]

19. (a) D H ✓
 (b) C G ✓
 (c) G ✓
 (d) G ✓
 (e) H ✓
 (f) A ✓
 (g) E ✓
 (h) A B ✓ [8]
20. (a) $(0.2 \times 0.75) + 0.8x = 0.39$ ✓
 $x = 0.3$ ✓
 (b) $\frac{0.15}{0.39} = \frac{5}{13}$ ✓✓
 (c) The events are not independent. ✓
 $P(S|B) \neq P(S)$
 $\frac{15}{39} \neq 0.2$ ✓ [6]
21. (a) (i) $V = \frac{4}{3} \pi (15)^3 = 4500\pi \text{ cm}^3$ ✓
 (ii) $\frac{4500\pi}{15^3} = k$ ✓
 $k = \frac{4}{3} \pi$
 (iii) $\tan \theta = \frac{4\pi}{3}$ ✓
 $\theta = 76.57^\circ$ ✓
- (b) Let x be the length, then $\frac{20}{x}$ is the width
 $(x + 3)\left(\frac{20}{x} + 1\right) = 40$ ✓
 $x = 5$ or 12 ✓
 Original dimensions are $4m \times 5m$ or $1\frac{2}{3}m \times 12m$ ✓
- (c) Domain: $\{x : x \neq 0, x \in \mathbb{R}\}$ ✓
 Range: $\{y : y \neq 1, y \in \mathbb{R}\}$ ✓ [9]
22. A: $y = \frac{1}{2} \sin x$ B: $y = 3 \sin x$ ✓✓
 The amplitude for a quieter sound is $\frac{1}{2}$.
 The amplitude increases to 3 to produce a louder sound. ✓
 C: $y = \sin \frac{x}{2}$ D: $y = \sin 4x$ ✓✓
 The period for a deeper pitch is 4π
 The period for a higher pitch is $\frac{\pi}{2}$. ✓ [6]